

Srujan Yamali

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science (GPA: 3.8/4.00)

Expected Graduation: May 2028

Relevant Coursework: Data Structures and Algorithms, Discrete Mathematics & Probability Theory, Structure & Interpretation of Computer Programs, Computer Architecture, Linear Algebra, Calculus I & II

EXPERIENCE

Global Clean Diesel

Cape Canaveral, FL

Software Development Intern

May 2026 – August 2026

- Wrote a 3D **compressible-flow solver (HLLC/MUSCL, SSP-RK2)** from scratch in **Python**, then rewrote the hot path in **Rust bit-for-bit identical** to it, scaling simulation from 3K to **51,468** channels in **15s** on 10 cores
- Automated a **66-case** parameter sweep that generated and executed **188** solver decks across **54** model tracks to size a **SpaceX 1.5 MW** genset, gated by **Monte-Carlo** uncertainty analysis and closed-form regression tests

Visa

Remote

Software Engineer Intern

January 2026 – May 2026

- Designed and deployed cloud-based **microservices** and **REST APIs** supporting Visa's risk and product organizations, reducing manual case verification time by **95%** and scaling to thousands of daily transactions
- Architected an **LLM-driven** document generation pipeline serving **1,600+** annual client implementations, adding caching layers that improved accuracy by **31%** and cut onboarding timelines by **45%**

Mercor

San Francisco, CA

Software Engineer

August 2025 – January 2026

- Shipped a config-driven **LLM evaluation pipeline** as a **FastAPI** service spanning **6** frontier models, replacing **15 min/record** serial grading with multi-key rotation and per-key **asyncio** semaphores to clear **700** runs in **6 hours**
- Engineered multi-provider **LLM orchestration** on AWS ECS with a **Redis**-backed distributed semaphore governing **200** concurrent calls, plus **MongoDB** and **SHA-256** content-hash caches that cut repeat processing by **87%**

Children's Hospital of Philadelphia

Philadelphia, PA

Data Science Intern

June 2024 – August 2025

- Rewrote the pairwise-similarity engine of an alignment-free **recombination detection** tool, replacing an **O(N²) NumPy** loop with a sparse **SciPy** matrix product that cut runtime **7,200s to 263s (27x)** over **2.44M** genomes
- Authored a **903-line** changepoint module segmenting N x N Jaccard matrices via **ruptures** BottomUp/L2 under a data-adaptive **BIC** penalty, adding **KDTree** k-NN pruning and vectorized **Mann-Whitney U** tests for **50x** faster scoring

Cornell University

Remote

Machine Learning Engineer Intern

September 2023 – May 2024

- Implemented **YOLO**-based object detection and multi-object tracking pipelines deployed on **Linux** servers, processing **500+** hours of real-world footage at **85%+** accuracy with multi-threaded inference optimization

PROJECTS

Stryda | [Site](#) | TypeScript, Node, Firebase Cloud Functions, Firestore, Claude API

Co-Founder

- Built the serverless backend behind an AI golf app: **36** callable endpoints, **9** HTTP routes, **8** event triggers and **4** cron jobs, fronting an agentic **tool-calling loop** that streams **Claude** turns with **GPT** fallback
- Settled every wager in a **deterministic engine** rather than the model, making offline score replay idempotent through composite-key upserts and gating writes behind role-based **security rules** and per-user daily **rate limits**

RedCarpet | [Github](#) | Python, Snakemake, LMDB, SciPy, Multiprocessing

First Author

- Shipped a **pip/conda**-installable CLI that orchestrates a five-stage **Snakemake** DAG over a **26 GB** sharded **LMDB** key-value store, fanned out across a **multiprocessing** worker pool and covered by **64** tests

SKILLS

Languages: Python, Go, C/C++, Java, TypeScript/JavaScript, Rust, SQL, MATLAB, Bash, HTML/CSS

Infrastructure: AWS (ECS, EC2, S3), GCP, Docker, Kubernetes, Firebase, Linux, PostgreSQL, MongoDB, Redis

Backend & AI/ML: FastAPI, Flask, Django, Node.js, Celery, PyTorch, TensorFlow, Scikit-Learn, OpenCV, LangChain

ACHIEVEMENTS

Finalist in Ramp's Builders Cup Hackathon; **3 conference publications** (IBANGS; Janelia Research Campus, HHMI; ISSSI); **Semi-finalist (~Top 10%)** USA Biology Olympiad; **1st Place**, 76th Delaware Valley Science Fair; **2nd Place BPA National** in Digital Communication, IT Concepts, Healthcare Administration; **1540 SAT**